

Machine Learning Applications

Winter semester 2025/2026



Lecture VII: Data Challenge Kick-Off

OVERVIEW

What is the idea behind the practical part of the exam?

- Work on an industry-related task with data from a real engineering application
 - Open task that requires creative thinking
 - **Apply machine learning / data mining techniques**
 - Work out your methods and results together in a group
 - Document your methodology and findings
 - Present your results in a short pitch
- ➔ Get a feeling for tasks on real data instead of solely academic tasks

What do we expect from you?

-  **Self-organize** your tasks and your group
-  Understand and get behind the problem from **industry perspective**
-  **Evaluate** your machine learning / data mining approaches
-  Review your results **critically**
-  **Justify** your proposed methods and algorithms
-  **Concise** documentation and presentation
-  Comprehensible programming code

Introduction

ETA RESEARCH FACTORY

Introduction

ETA Research Factory

- PTW | Institute for Production Management, Technology and Machine Tools
- ETA | Energy Technologies and Applications in Production
- The ETA Research Factory offers a representative production chain of the metalworking industry



Pictures: PTW TU Darmstadt

Introduction

ETA Research Factory

TEC



Manufacturing
Technology

CiP



Center for Industrial
Productivity

**Production
Organization**

Prof. Dr.-Ing. J. Metternich

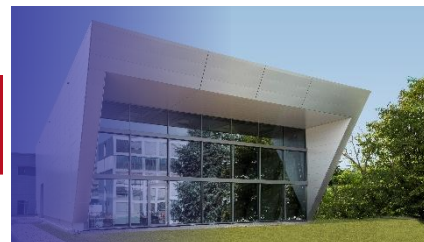
**Data-driven
Production**

Prof. Dr.-Ing. M. Weigold

**Climate-neutral
Production**

Prof. Dr.-Ing. M. Weigold

MiP



Management of
Industrial Production

ETA



Energy Technologies
and Application in
Production

Introduction

ETA Research Factory



Energy
Efficiency



Energy
Flexibility



Ressource
Efficiency

The research group ETA (Energy Technologies and Applications in Production) strives for the vision of making tomorrow's industrial production energy-efficient, energy-flexible and resource-efficient, and thus making a significant contribution to climate neutral production.



Climate-neutral production

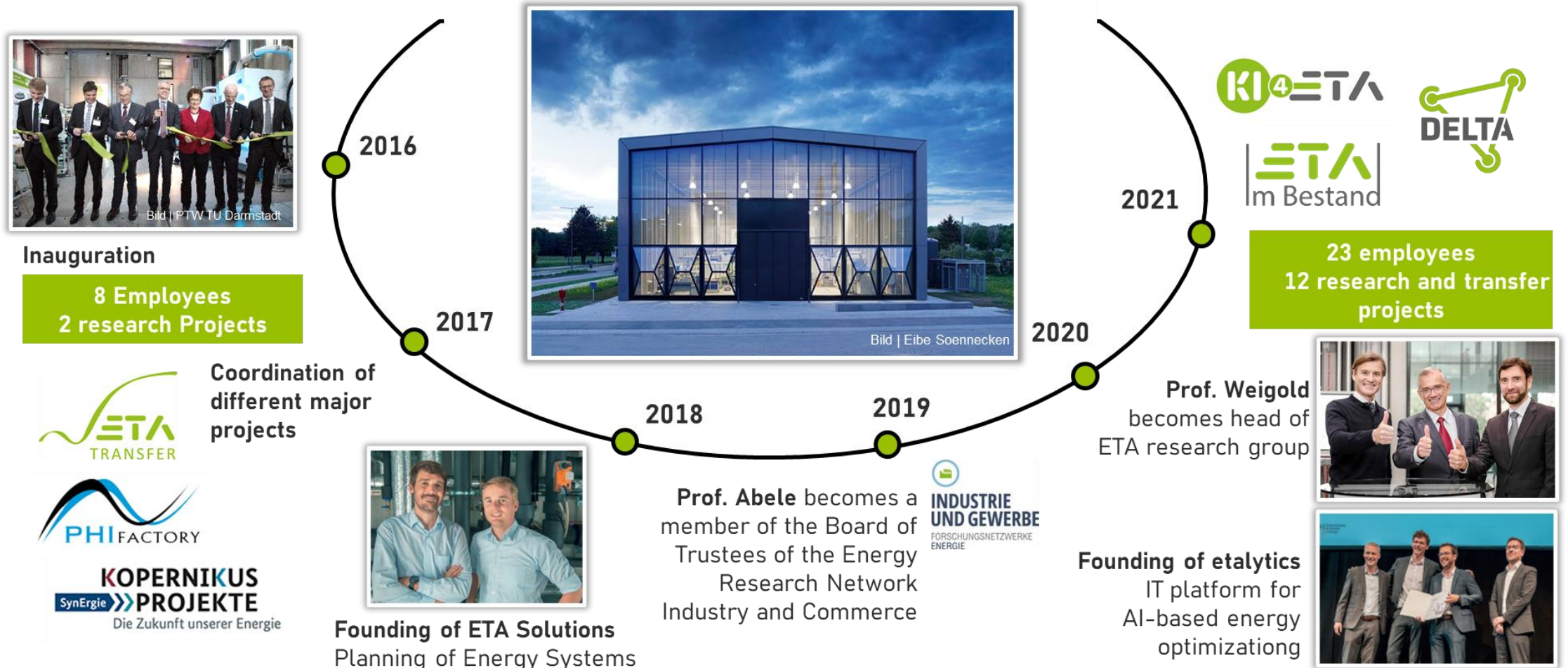
Focus: Direct and indirect emissions in factory operation



Source | Eibe Soennecken

Introduction

ETA Research Factory



Introduction

ETA Research Factory

Working fields: Energy efficiency // Energy flexibility // Resource efficiency



Strategic Energy- and Ressource Management

Transformation strategies •
Transfer, Competences &
Learning Factories

How can companies
strategically align their
production with the goal of
climate neutrality?

Bild | pixabay



Climate Neutral Production Infrastructure

Cooling • Heat • Compressed
Air • Ventilation

How do you plan and operate
climate-neutral production
infrastructure?

Bild | stock.adobe



Energetic System Analysis and Optimization of Production Machines

Machining • Cleaning • Drying •
Heat Treatment

How can efficiency and
flexibility potentials on
production machines be
investigated and exploited?

Bild | Jari Hosan



Energy-optimized Factory Operation

Control Strategies •
Autonomous Operation •
Forecasting

How can optimization of
factory operations contribute
to climate-neutral
production?

Bild | stock.adobe



Cyber-physical Systems in Climate-neutral Production

CPPS • Digital Twin • Asset
Administration Shell

How must which data be
structured, communicated
and standardized to enable
climate-neutral production?

Bild | stock.adobe

Goal: Development of technologies, solutions and methods
for climate-neutral production

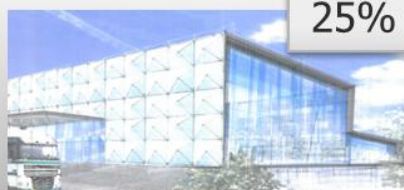
Methods: Experimental setups // Data models // Simulation // Optimization // Artificial intelligence

Introduction

ETA Research Factory

Status Quo: Isolated research on separate factory sub-systems

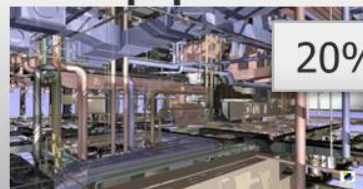
Sub-system building construction



25%

Quelle: Prof. Dipl.-Ing. J. Eisele

Sub-system technical building equipment



20%

Sub-system machine



30%

Total
savings
< 30 %

Our goal: Holistic factory optimization considering all sub-systems



Interaction of:

- Machines
- Technical building equipment
- Building construction

**Synergies through crosslinking,
energy controlling and recuperation**

Total
potential
~ 40 %

Introduction

ETA Research Factory

Machine

EMAG VLC 100Y
Vertical lathe machine



MAFAC JAVA
Cleaning machine



IVA
Rotary Retort Furnace



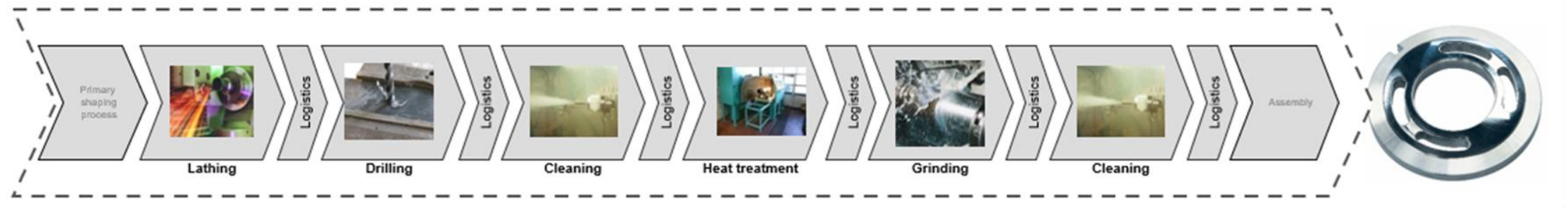
EMAG VLC 100 GT
Vertical grinding machine



MAFAC KEA
Cleaning machine



Control plate for axial
piston-hydraulics

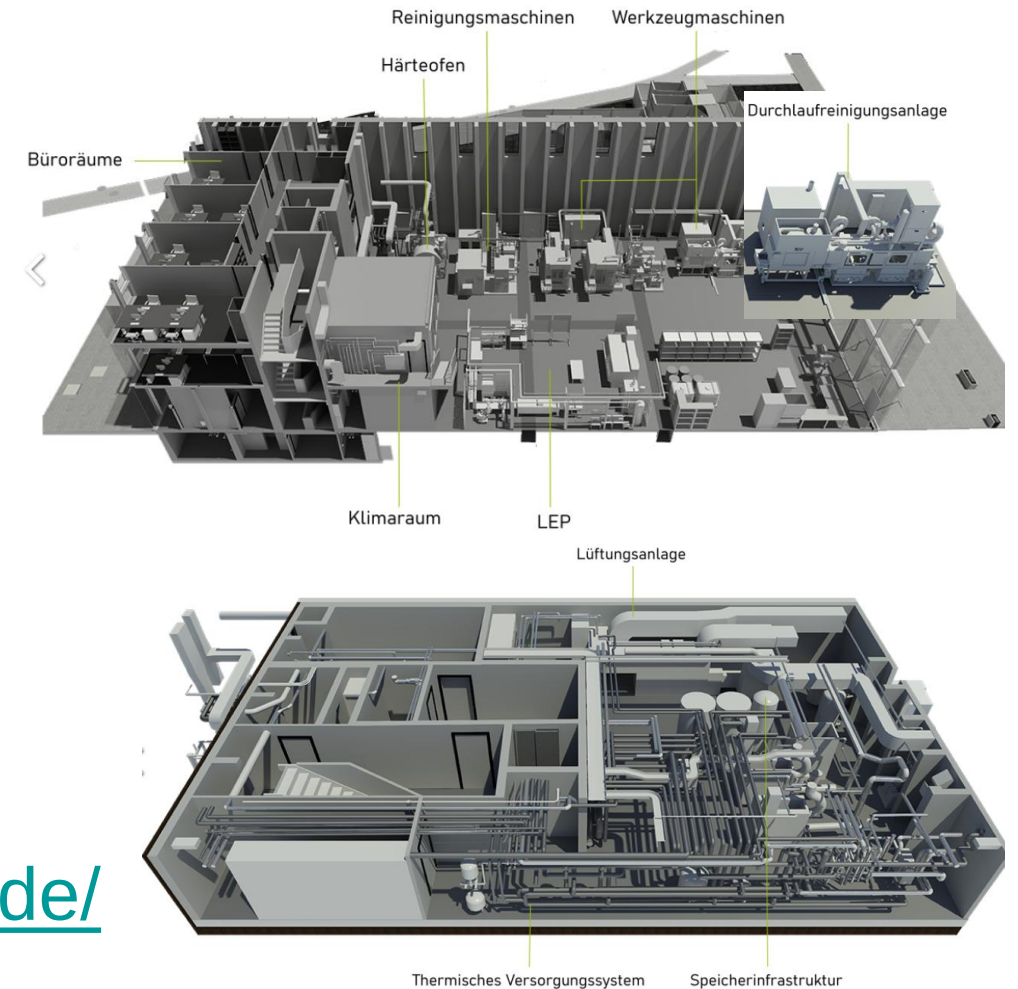


Real production process chain
Innovative production building
Interdisciplinary approach

Introduction

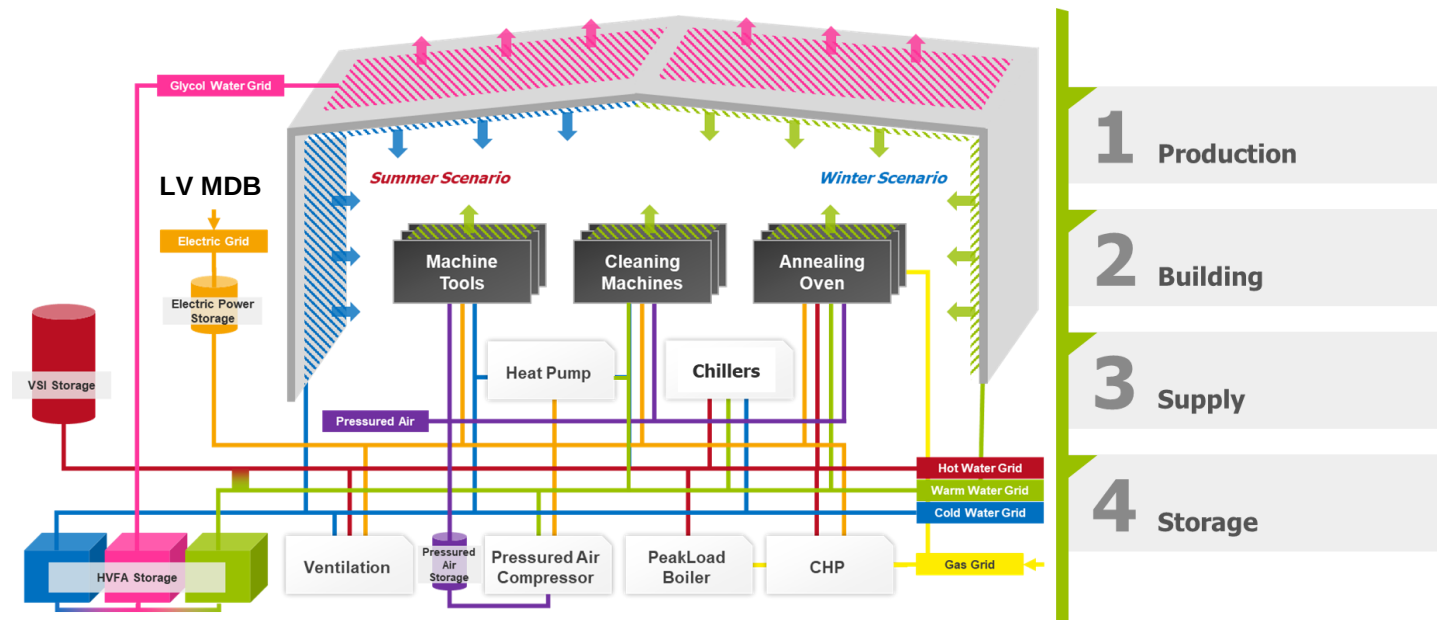
ETA Research Factory

- 770 m² floor area
- Real process chain with:
 - Machining
 - Hardening
 - Cleaning
 - Drying
- Energy supply system
- <https://www.360grad.ptw.tu-darmstadt.de/>



https://www.ptw.tu-darmstadt.de/eta-fabrik/etafabrik/ausstattung_eta/index.de.jsp

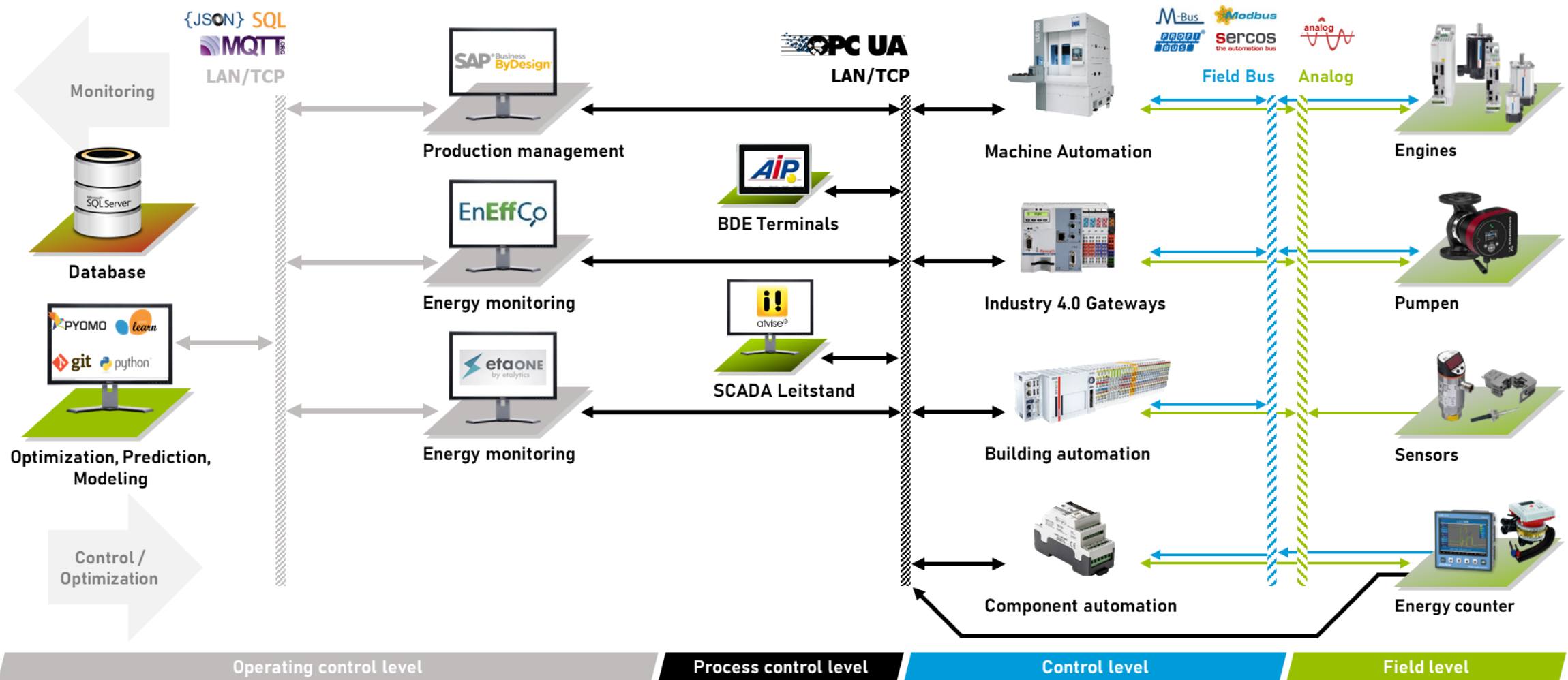
- Main point of power distribution into the industrial facility: Low Voltage Main Distribution Board (LV MDB)
- Ensures supply of electrical energy to all production lines and auxiliary systems



Picture: PTW TU Darmstadt

Introduction

ETA Research Factory



Introduction

ETA Research Factory

The ETA Research Group regularly offers opportunities for **theses (Bachelor, Master, Projects)**, **student research assistants**, and **research associate** positions.

If you are interested in contributing to our ongoing projects and advancing your academic or professional career, **feel free to contact us** — we look forward to hearing from you!

Jobs → https://www.ptw.tu-darmstadt.de/institut_ptw/karriere_ptw/index.en.jsp

Theses → https://www.ptw.tu-darmstadt.de/studium/abschlussarbeiten_ptw/index.en.jsp



M. Sc. Robin Zink
Research Associate
Institut für Produktionsmanagement,
Technologie und Werkzeugmaschinen
L6|03 R203
Eugen-Kogon-Straße 4
64287 Darmstadt
+49 6151 82 296 38
r.zink@ptw.tu-Darmstadt.de



Prof. Dr.-Ing. Matthias Weigold
Institute Director

MLA 2025/26

DATA CHALLENGE TOPIC

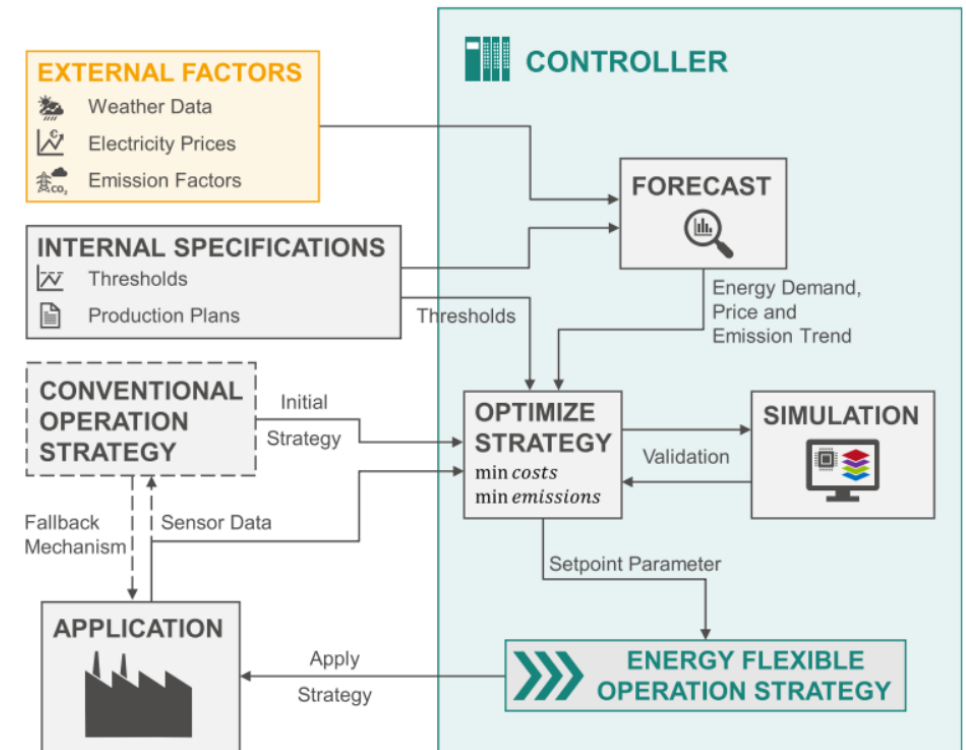
LOW VOLTAGE MAIN DISTRIBUTION BOARD (LV MDB) DATA OF ETA RESEARCH FACTORY



Picture: PTW TU Darmstadt

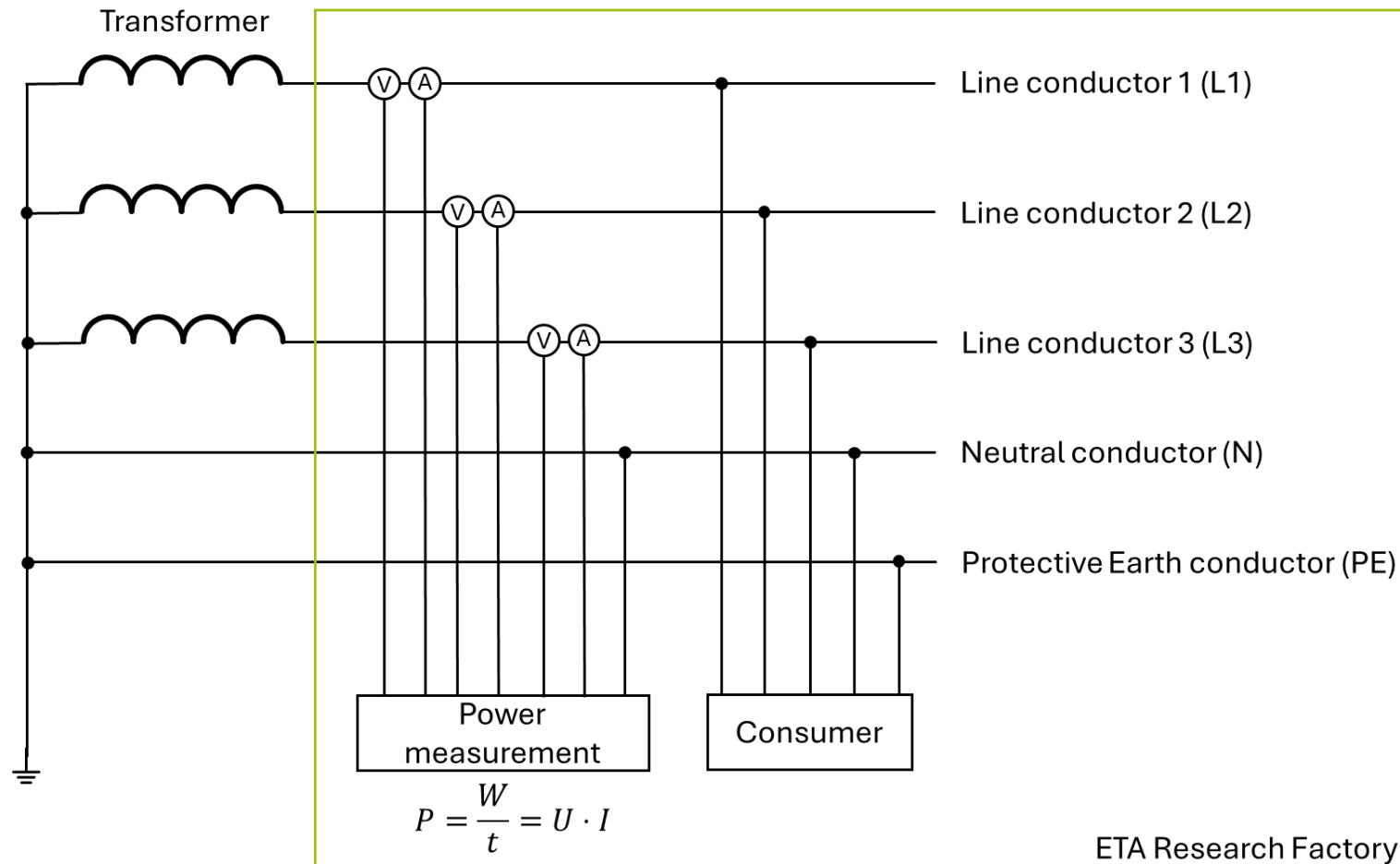
Challenge for Energy System Operators

- Energy consumption is critical for industrial operations
 - Fundamental **enabler** of production processes (e.g., machining, heating, compressed air, automation).
 - Contributes significantly to the **total costs**.
- Optimizing industrial energy use is highly complex
 - Production systems involve **diverse** loads (base load, variable process loads, auxiliary systems) with strong interdependencies.
 - Energy optimization must **balance costs and reliability**.
- Precise load forecasts are an enabler for optimization
 - Accurate forecasts enable scheduling, peak load management, and integration of renewables or storage systems.



Wendt, J.; Zangenberg, J.; Weigold, M.: Optimized Energy-Flexible Operation Strategies of Air Conditioning Systems in Production Environments. 2020

LV MDB Data - ETA Research Factory

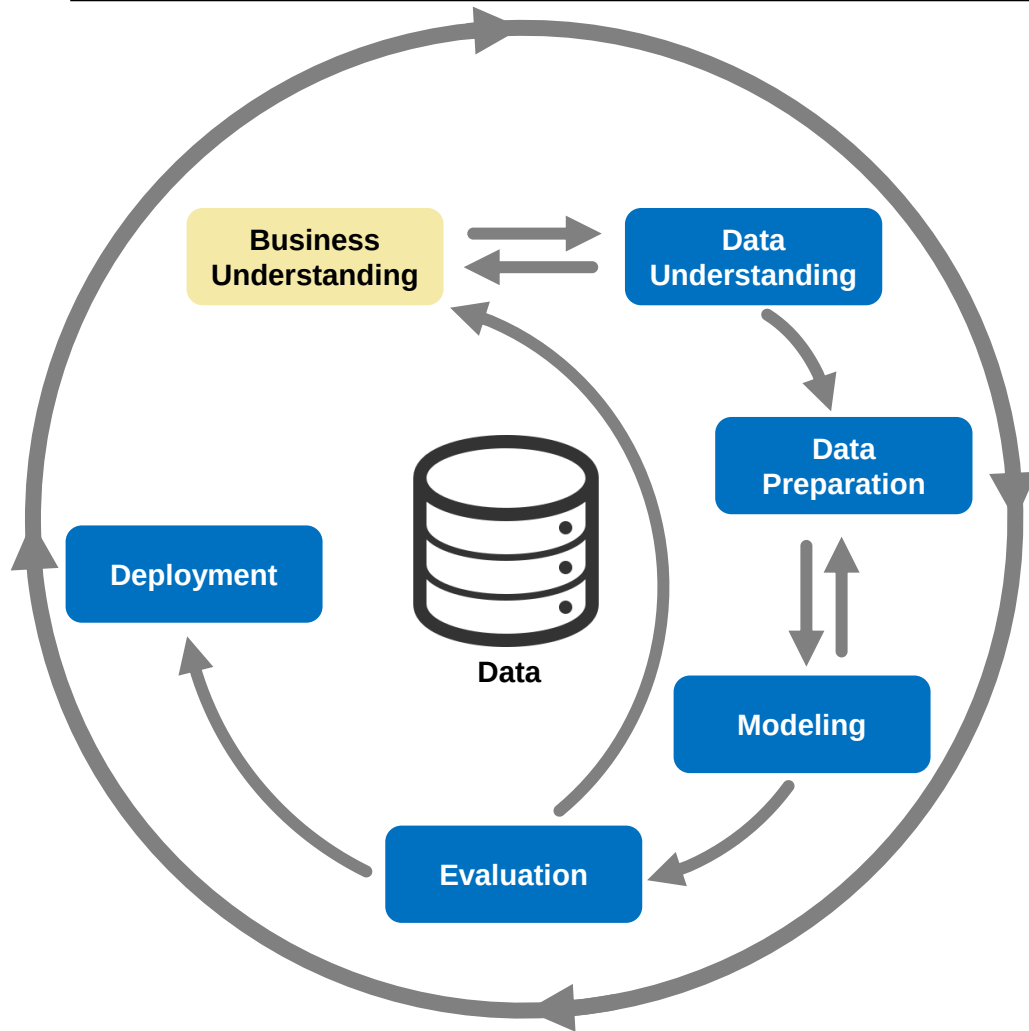


Legend:

- P ... Electric power, in watt
- W ... Electric work/energy, in joule
- t ... Time, in seconds
- U ... Electric voltage, in volt
- I ... Electric current, in ampere

Possible Approach: CRISP-DM

Cross-Industry Standard Process for Data Mining



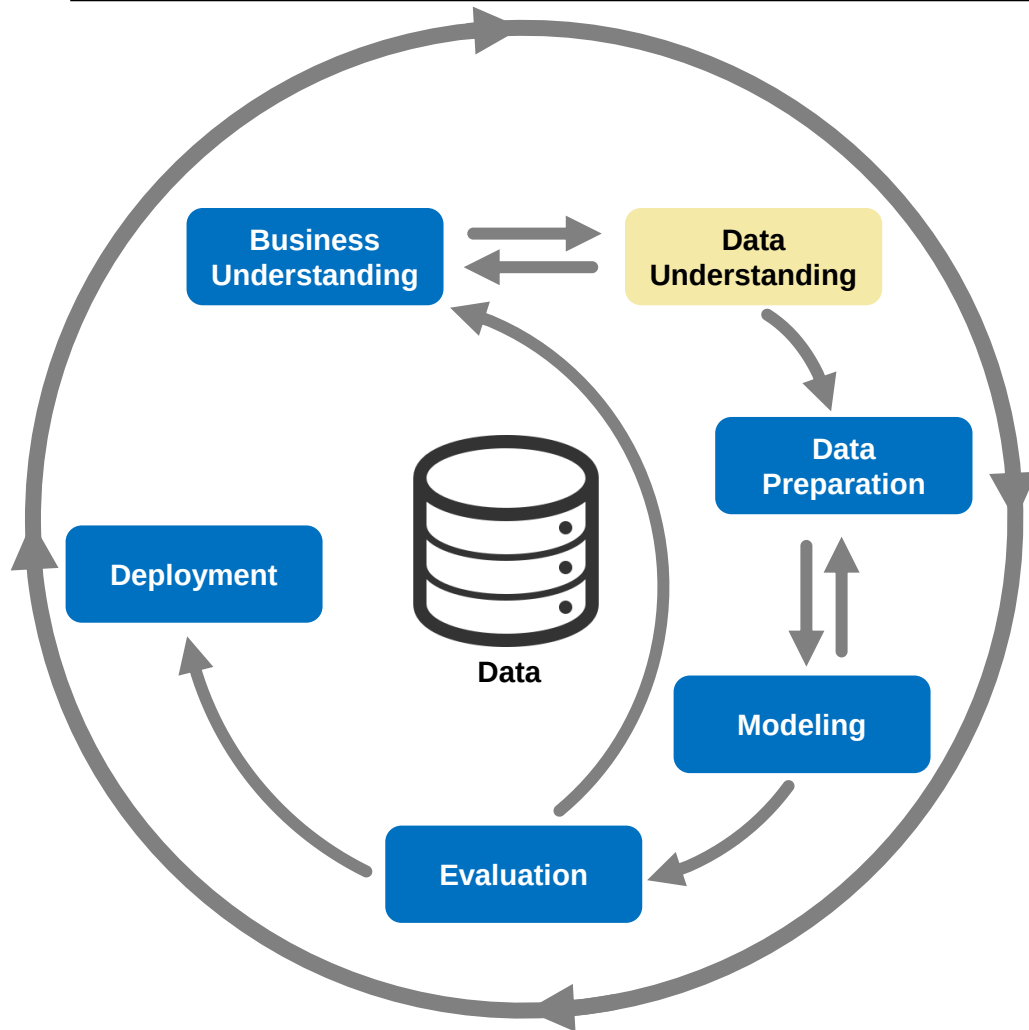
- **Iterative open standard process** model to gain knowledge from data related to business goals
- **Feedback loops** allow iteration of goals and models

1. **Define project goals and business objectives**
2. Understand the available data and their quality
3. Filter and select useful and relevant data
4. Create data-models that might meet the defined goals
5. Evaluate models' performance related to the goals
6. Set the best model into operation

Source: <ftp://public.dhe.ibm.com/software/analytics/spss/documentation/modeler/18.0/en/ModelerCRISPDm.pdf>

Possible Approach: CRISP-DM

Cross-Industry Standard Process for Data Mining

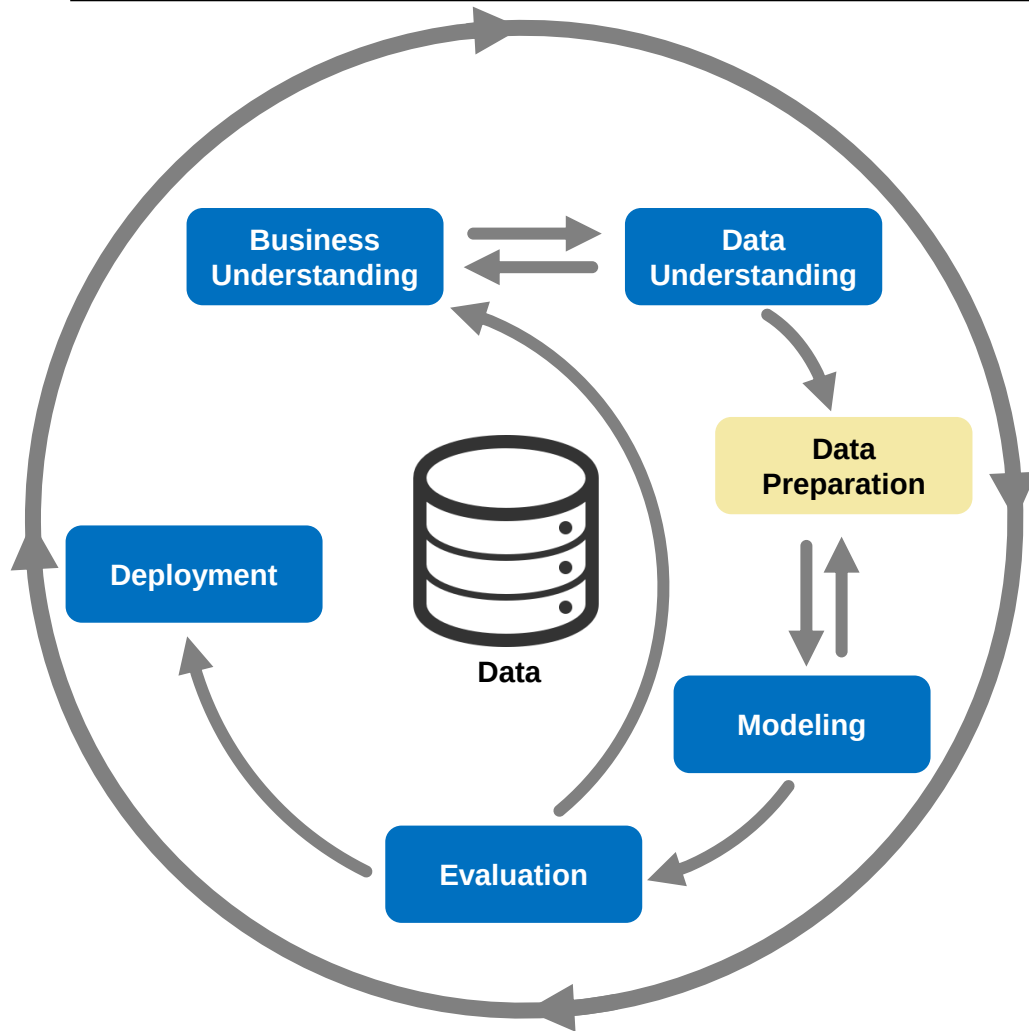


- **Iterative open standard process** model to gain knowledge from data related to business goals
 - **Feedback loops** allow iteration of goals and models
1. Define project goals and business objectives
 2. **Understand the available data and their quality**
 3. Filter and select useful and relevant data
 4. Create data-models that might meet the defined goals
 5. Evaluate models' performance related to the goals
 6. Set the best model into operation

Source: <ftp://public.dhe.ibm.com/software/analytics/spss/documentation/modeler/18.0/en/ModelerCRISPDm.pdf>

Possible Approach: CRISP-DM

Cross-Industry Standard Process for Data Mining

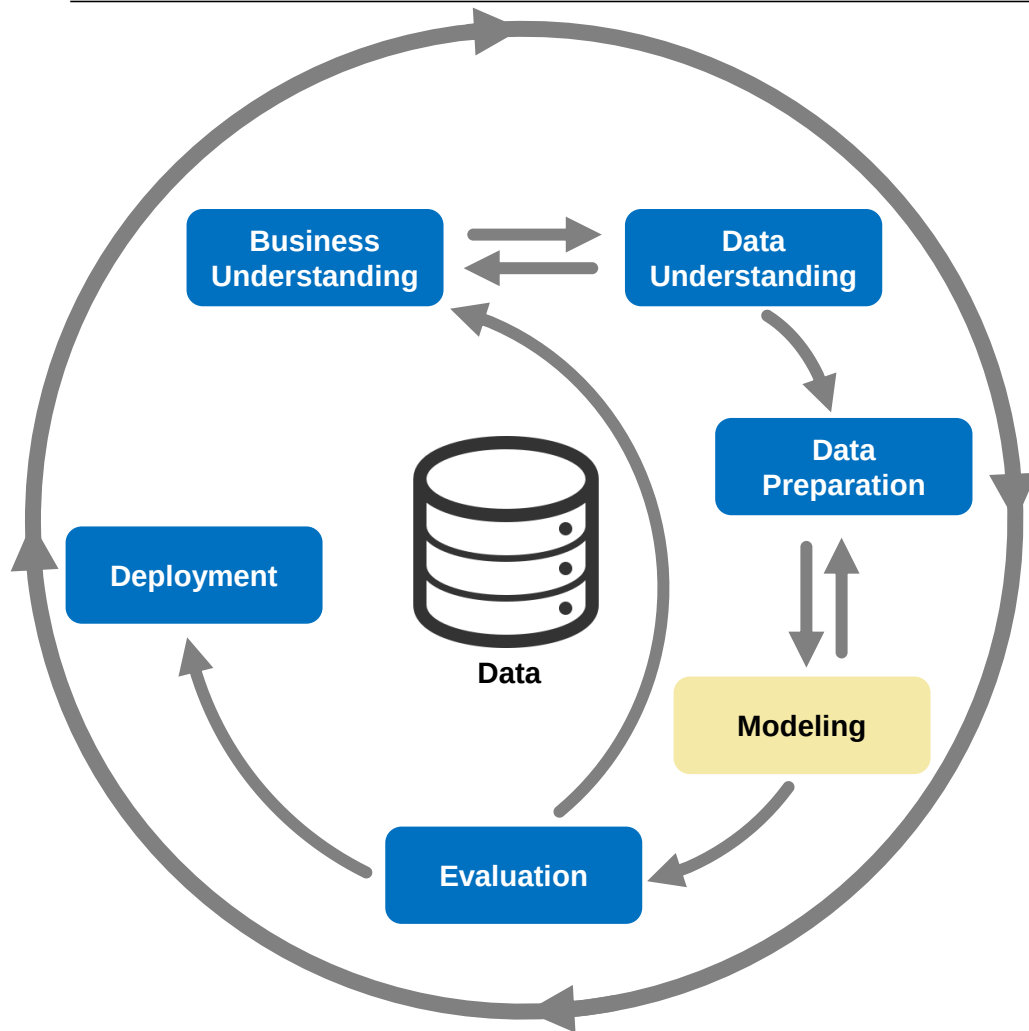


- **Iterative open standard process** model to gain knowledge from data related to business goals
 - **Feedback loops** allow iteration of goals and models
1. Define project goals and business objectives
 2. Understand the available data and their quality
 3. **Filter and select useful and relevant data**
 4. Create data-models that might meet the defined goals
 5. Evaluate models' performance related to the goals
 6. Set the best model into operation

Source: <ftp://public.dhe.ibm.com/software/analytics/spss/documentation/modeler/18.0/en/ModelerCRISPDM.pdf>

Possible Approach: CRISP-DM

Cross-Industry Standard Process for Data Mining



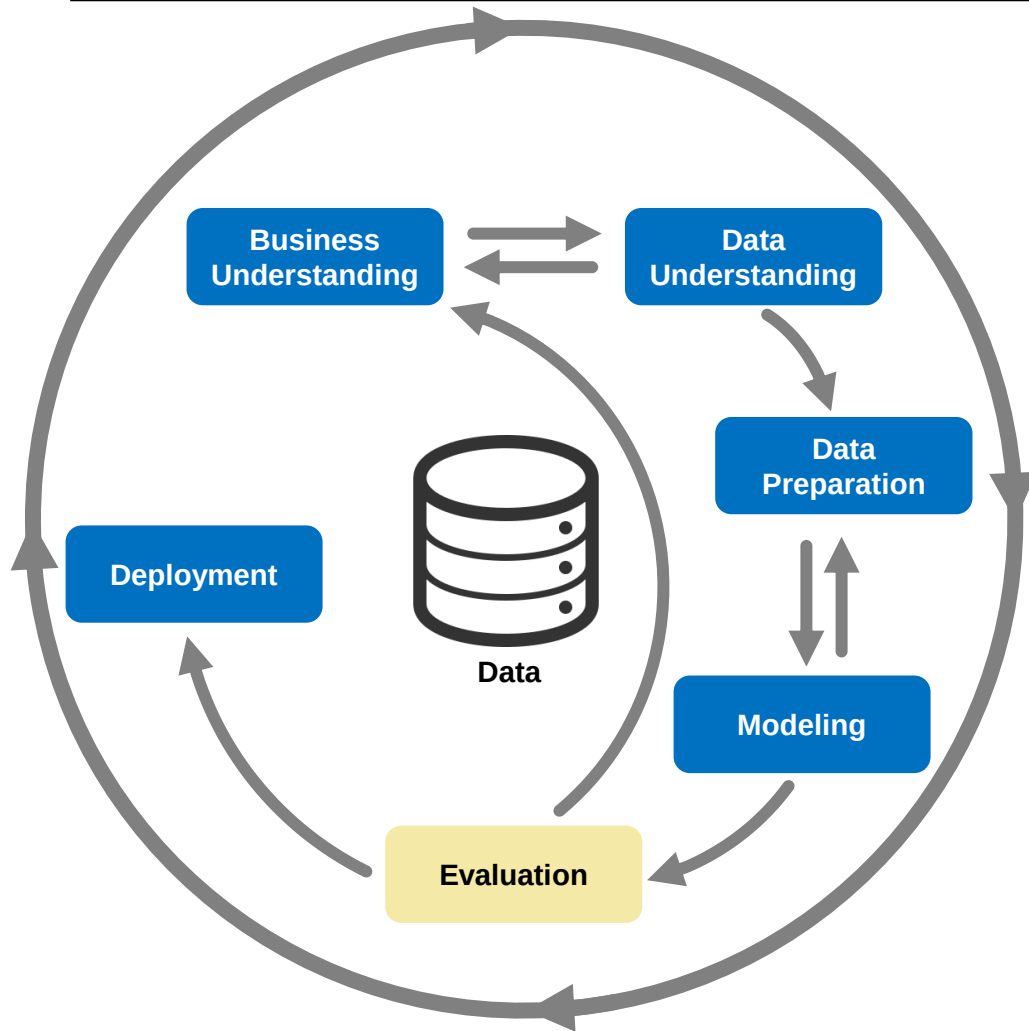
- **Iterative open standard process** model to gain knowledge from data related to business goals
- **Feedback loops** allow iteration of goals and models

1. Define project goals and business objectives
2. Understand the available data and their quality
3. Filter and select useful and relevant data
4. **Create data-models that might meet the defined goals**
5. Evaluate models' performance related to the goals
6. Set the best model into operation

Source: <ftp://public.dhe.ibm.com/software/analytics/spss/documentation/modeler/18.0/en/ModelerCRISPDM.pdf>

Possible Approach: CRISP-DM

Cross-Industry Standard Process for Data Mining

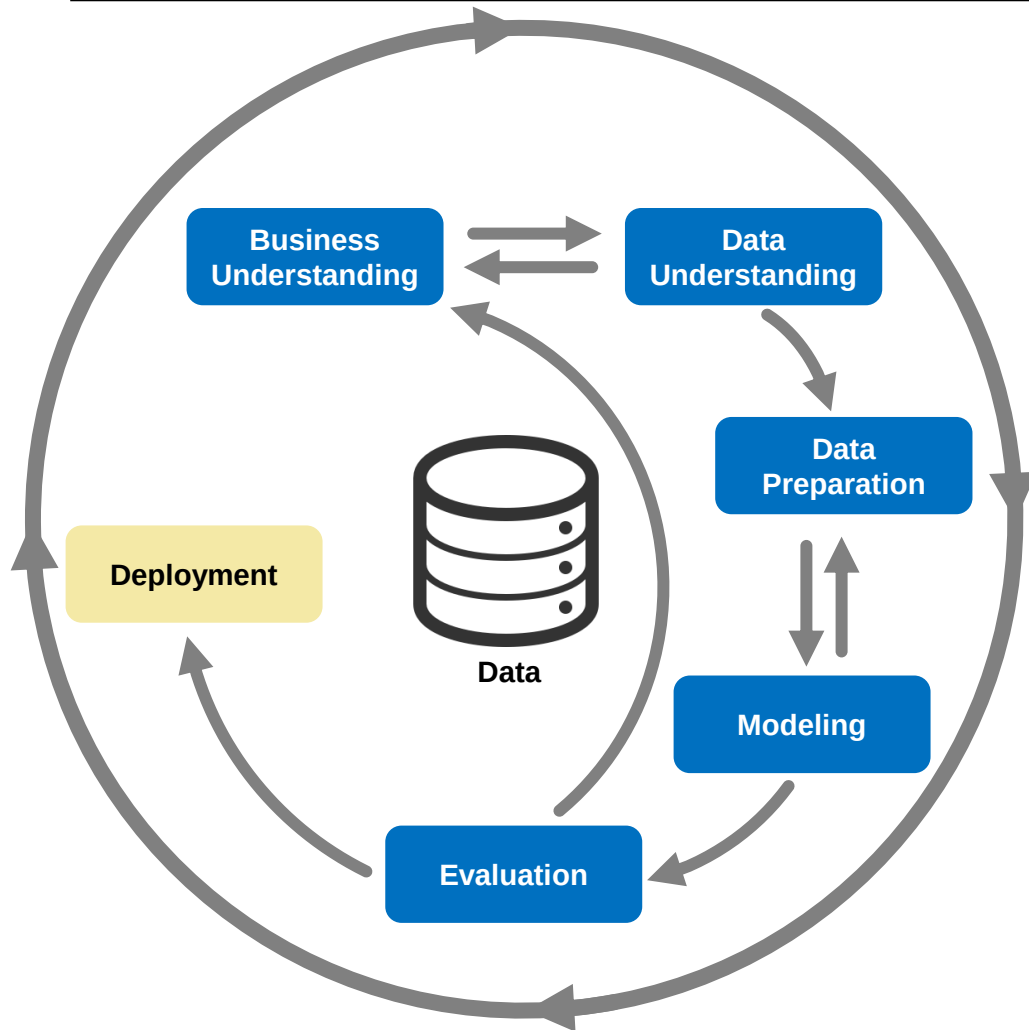


- **Iterative open standard process** model to gain knowledge from data related to business goals
 - **Feedback loops** allow iteration of goals and models
1. Define project goals and business objectives
 2. Understand the available data and their quality
 3. Filter and select useful and relevant data
 4. Create data-models that might meet the defined goals
 5. **Evaluate models' performance related to the goals**
 6. Set the best model into operation

Source: <ftp://public.dhe.ibm.com/software/analytics/spss/documentation/modeler/18.0/en/ModelerCRISPDM.pdf>

Possible Approach: CRISP-DM

Cross-Industry Standard Process for Data Mining



- **Iterative open standard process** model to gain knowledge from data related to business goals
 - **Feedback loops** allow iteration of goals and models
1. Define project goals and business objectives
 2. Understand the available data and their quality
 3. Filter and select useful and relevant data
 4. Create data-models that might meet the defined goals
 5. Evaluate models' performance related to the goals
 6. **Set the best model into operation**

Source: <ftp://public.dhe.ibm.com/software/analytics/spss/documentation/modeler/18.0/en/ModelerCRISPDM.pdf>

Task:

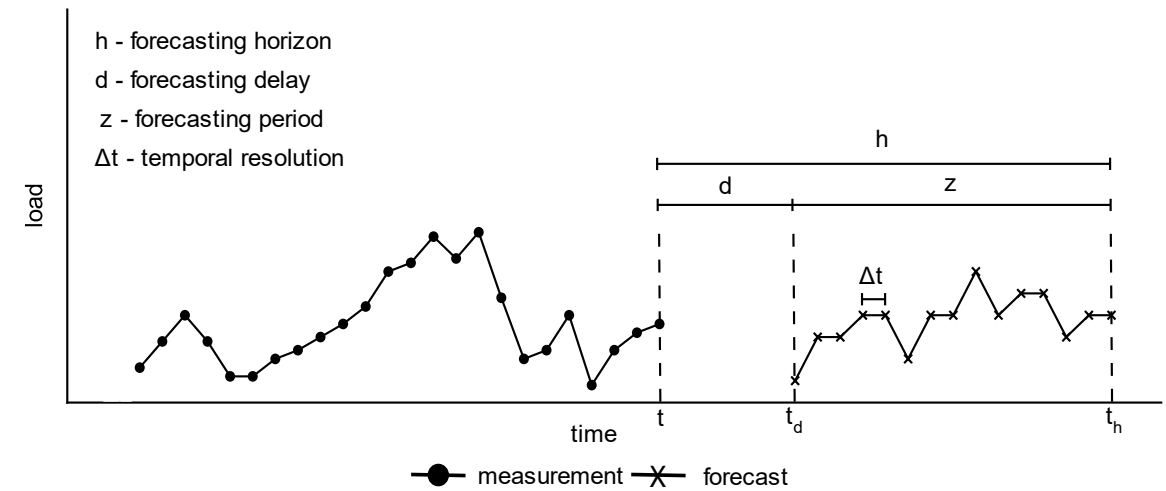
Electrical load forecast of the **LV MDB - ETA Research Factory** (kW) at each time step for the next 24 hours ($h = z = 24$ h) in hourly time resolution ($\Delta t = 1$ h), i.e. 24 targets per forecasting step.

Target error metric:

Mean absolute error (**MAE**) per horizon.

Features:

- Power measurement features
- Datetime features
- Weather features



Based on source: <https://tuprints.ulb.tu-darmstadt.de/21767/>

Data Understanding: First Overview

LV MDB Data - ETA Research Factory

Feature category	Code	Description
Power measurement	Elec_P_LVMDB_ETA_kW	Electrical power LV MDB ETA Factory in kW
Datetime	Time	Datetime in format yyyy-mm-dd hh:mm:ss
Weather	humidity_percent	Humidity in %
	precipitation_height_mm	Precipitation height in mm
	temperature_air_mean_C	Mean air temperature in °C
	wind_speed_m_per_s	Wind speed in m/s

→ In addition to the existing features, further features can be acquired from public sources and new features may be engineered!

Electrical load forecast of the **LV MDB - ETA Research Factory** (kW) for the next 24 hours (**$h = z = 24 \text{ h}$**) in hourly time resolution (**$\Delta t = 1 \text{ h}$**).

Inputs: time series data with

- LV MDB power consumption
- Weather features
- Datetime features of the ETA Research Factory



ML model(s)



Output: forecast next 24 hourly

- LV MDB power consumptions of the ETA Research Factory

Evaluation metric:

MAE of LV MDB power consumption at specified forecasting horizons

Target preparation for forecasting

Slide is from lecture 10

The **forecasting horizon** is the future time period for which a forecast is generated. For instance, if forecasting the next 100 seconds into the future, the forecasting horizon is 100 seconds.

Initial data set

Time (s)	Feature 1	Feature 2	Electric load
-	-	-	-
0	$x_1^{(0)}$	$x_2^{(0)}$	$y^{(0)}$
1	$x_1^{(1)}$	$x_2^{(1)}$	$y^{(1)}$
2	$x_1^{(2)}$	$x_2^{(2)}$	$y^{(2)}$
3	$x_1^{(3)}$	$x_2^{(3)}$	$y^{(3)}$
...	...	...	...

Prepared data set

Time (s)	Feature 1	Feature 2	Electric load	Electric load (shifted)
-	-	-	-	$y^{(0)}$
0	$x_1^{(0)}$	$x_2^{(0)}$	$y^{(0)}$	$y^{(1)}$
1	$x_1^{(1)}$	$x_2^{(1)}$	$y^{(1)}$	$y^{(2)}$
2	$x_1^{(2)}$	$x_2^{(2)}$	$y^{(2)}$	$y^{(3)}$
3	$x_1^{(3)}$	$x_2^{(3)}$	$y^{(3)}$	...
...	...	...	...	...

Feature engineering for forecasting

Slide is from lecture 10

D **Temporal features** are features that capture the time related dependencies in a time series model. Examples for temporal features in forecasting tasks are time lag features and moving average features.

Time (s)	Feature 1	Time lag feature 1 [1 step]	Feature 2	Moving average feature 2 [window size 2]
0	$x_1^{(0)}$	-	$x_2^{(0)}$	-
1	$x_1^{(1)}$	$x_1^{(0)}$	$x_2^{(1)}$	$\text{mean}(x_2^{(0)}, x_2^{(1)})$
2	$x_1^{(2)}$	$x_1^{(1)}$	$x_2^{(2)}$	$\text{mean}(x_2^{(1)}, x_2^{(2)})$
3	$x_1^{(3)}$	$x_1^{(2)}$	$x_2^{(3)}$	$\text{mean}(x_2^{(2)}, x_2^{(3)})$
4	...	$x_1^{(3)}$	$x_2^{(4)}$	$\text{mean}(x_2^{(3)}, x_2^{(4)})$
...	...	...	...	...
n	$x_1^{(n)}$	$x_1^{(n-1)}$	$x_2^{(n)}$	$\text{mean}(x_2^{(n-1)}, x_2^{(n)})$



REGISTRATION PROCEDURE

→ The process is also described in **Moodle** and, more detailed, in the document “**Notes on the examination**”! Please read the “**IT-Terms of Use**” as well!

Two-step registration process

Exam Registration and Group Selection

1. Registration in TUCaN



- Register in TUCaN for the examination “*Machine Learning Applications (Group Work)*”

Start: Today
Deadline: Sun, December 7

After registration deadline in TUCaN, you must accomplish the exam!

2. Group selection process in Moodle



- Only available after registration in TUCaN (updated once per day)
- Choose your favourite group
 - Targeted group size: **4 to 5 students**
 - Students who do not have a preferred group, can use the option “No group choice”
 - They will randomly be assigned to a group after the end of the group selection period

Start: Thu, December 4
Deadline: Mon, December 8

 **GROUP CHOICE**
Registration for your group 

You can select your desired group and form a group with your preferred fellow students. The target group size is four to five people. You can change your group choice on your own throughout the selection period.

If you do not have a preferred group, please use the option "No group choice". We will then randomly assign you to a group after the end of the group selection period.

Period: 6.12.2023 10:00 till 10.12.2023 23:59

 Not available unless:

- You belong to **Examinees** (hidden otherwise)
- It is after **6 December 2023, 10:00 AM**
- It is before end of **10 December 2023**

Start of the Data Challenge

Working on the data challenge task

- You have to work on the data challenge task on **your own hardware!**

After successful registration, we will send an e-mail to your student mail address on December 9th with information regarding

- Access to the data
- Additional IT-Services for the data challenge that you can, but do not have to use



File Server

- Data storage
- Data synchronisation



Gitlab

- Code versioning
- Collaborative code work



Jupyterhub Server

- Computational resources (limited)
- Programming environment

Start: Tue, December 10
End: Mon, February 16

End of the data challenge

Deliverables

1. Documentation

- Use Jupyter-Notebook template with given structure
 - Developed **programming code**
 - Use text cells for **documentation**
- Hand-in your notebook latest on February 16, 2026

2. Presentation

- Use available PowerPoint-Template as orientation
- You have **maximum 10 minutes** to present and approximately **5 minutes** for discussion
- All members of the group need to attend at the given appointment
- Presentation dates will be published soon

3. Test set predictions of your machine learning model

- Your final machine learning model will be evaluated on a hold-out test set (target not available to students)
- Hand-in your final predictions latest on February 16, 2026

➔ All obligatory details can be found in **Notes on the examination** in Moodle

Assistance during the Data Challenge

- Moodle discussion platform can be used among the students
 - Further information on the data can be found in Moodle and on the file server
- Please use this service before reaching out to the research associates

- **Weekly consultation hour** with research associates:
 - Tuesdays, 09:00 AM – 10:00 AM in L1|01-K562



Summary and further announcements

- In order to participate, register in TuCan **before the deadline!**
- Be prepared to receive mails via student-Account



Next week: Lecture VIII – Classification with support vector machines

Thank you for your attention



M. Sc. Robin Zink
Research Associate
Institut für Produktionsmanagement,
Technologie und Werkzeugmaschinen
L6|03 R203
Eugen-Kogon-Straße 4
64287 Darmstadt
+49 6151 82 296 38
r.zink@ptw.tu-Darmstadt.de



Prof. Dr.-Ing. Matthias Weigold
Institute Director

